

Workshop 4 2024.11.07

At which points are functions continuous?

23. $y = \frac{x \tan x}{x^2 + 1}$

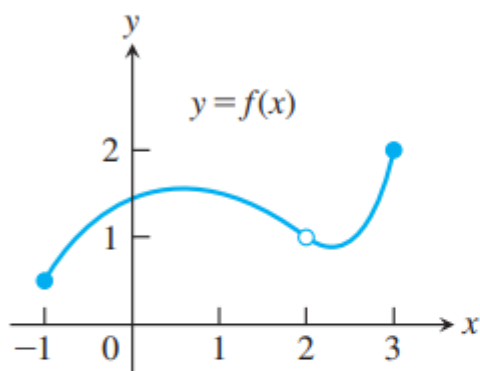
24. $y = \frac{\sqrt{x^4 + 1}}{1 + \sin^2 x}$

30. $f(x) = \begin{cases} \frac{x^3 - 8}{x^2 - 4}, & x \neq 2, x \neq -2 \\ 3, & x = 2 \\ 4, & x = -2 \end{cases}$

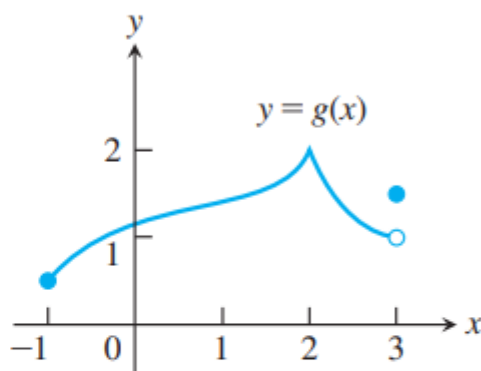
Continuity from Graphs

In Exercises 1–4, say whether the function graphed is continuous on $[-1, 3]$. If not, where does it fail to be continuous and why?

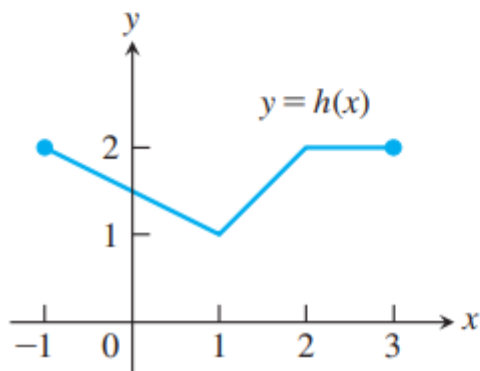
1.



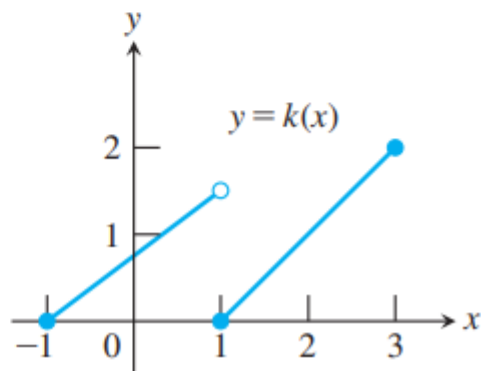
2.



3.



4.



46. For what values of a and b is

$$g(x) = \begin{cases} ax + 2b, & x \leq 0 \\ x^2 + 3a - b, & 0 < x \leq 2 \\ 3x - 5, & x > 2 \end{cases}$$

continuous at every x ?

37. Define $g(3)$ in a way that extends $g(x) = (x^2 - 9)/(x - 3)$ to be continuous at $x = 3$.

Find limits at $+\infty$ and at $-\infty$

7. $h(x) = \frac{-5 + (7/x)}{3 - (1/x^2)}$

8. $h(x) = \frac{3 - (2/x)}{4 + (\sqrt{2}/x^2)}$

Find the limits in Exercises 9–12.

9. $\lim_{x \rightarrow \infty} \frac{\sin 2x}{x}$

10. $\lim_{\theta \rightarrow -\infty} \frac{\cos \theta}{3\theta}$

11. $\lim_{t \rightarrow -\infty} \frac{2 - t + \sin t}{t + \cos t}$

12. $\lim_{r \rightarrow \infty} \frac{r + \sin r}{2r + 7 - 5 \sin r}$

Domains, Ranges, and Asymptotes

Determine the domain and range of each function. Use various limits to find the asymptotes and the ranges.

69. $y = 4 + \frac{3x^2}{x^2 + 1}$

70. $y = \frac{2x}{x^2 - 1}$

71. $y = \frac{\sqrt{x^2 + 4}}{x}$

72. $y = \frac{x^3}{x^3 - 8}$

Oblique Asymptotes

Graph the rational functions in Exercises 103–108. Include the graphs and equations of the asymptotes.

103. $y = \frac{x^2}{x - 1}$

104. $y = \frac{x^2 + 1}{x - 1}$

105. $y = \frac{x^2 - 4}{x - 1}$

106. $y = \frac{x^2 - 1}{2x + 4}$

107. $y = \frac{x^2 - 1}{x}$

108. $y = \frac{x^3 + 1}{x^2}$

Use formal definitions to prove the limit statements in Exercises 93–96.

93. $\lim_{x \rightarrow 0} \frac{-1}{x^2} = -\infty$

94. $\lim_{x \rightarrow 0} \frac{1}{|x|} = \infty$

95. $\lim_{x \rightarrow 3} \frac{-2}{(x - 3)^2} = -\infty$

96. $\lim_{x \rightarrow -5} \frac{1}{(x + 5)^2} = \infty$